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Paleomagnetism of Permo-Triassic Redbeds from the Asturias and Cantabric Chain (northern Spain): evidence for strong lower Tertiary remagnetizations

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Abstract

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The paleomagnetic analysis of the Permo-Triassic redbeds outcropping in the western part of the Cantabric Chain and the small Mesozoic basin from the Asturias shows that these formations have a history of complex magnetization.

Only a few sites did not experience the remagnetization processes and retained original directions. The most reliable results yield a paleomagnetic pole located at: lat. 49°N; long. 217°E ($n = 11$, $\alpha_{95} = 3.7°$), which is suggested as reliable Permo-Triassic data for the Iberian plate. Two remagnetization phases are recognized: a moderate phase predating the folding gave rise to a first overprinting. It is connected with the distension which occurred in the Pyreneo-Cantabrian region during the upper Jurassic–lower Cretaceous. The main remagnetization phase which occurred after the folding is dated from the lower Tertiary, and can be related to the compression induced on the northern boundary of Iberia from upper Cretaceous onwards. In some cases this phase led to a complete replacement of the primary magnetization.

Previously published data, which were at the time interpreted as being European-like in direction, are attributed to this phase. Hence, our results do not support the hypothesis of a micro-plate called "le Danois block", which was suggested in order to explain these results. We believe that there is no paleomagnetic evidence supporting the existence of a complicated boundary between Europe and Iberia during the mid-Cretaceous opening of the Bay of Biscay.

1. Introduction

Marine geological and geophysical data collected in the Bay of Biscay has shown that the Spanish northern margin is an extension of the Pyrenees (e.g. Le Pichon and Sibuet, 1971; Boillot et al., 1979; Boillot, 1984), and was deformed mainly during the upper Cretaceous–Eocene compressional phase. Although the kinematics of Iberian relative to Europe prior to that time is still a matter of debate (e.g., Olivet and al., 1984; Masson and Miles, 1984), the models generally assume that the northern margin of Spain belongs to the Iberian plate. This idea was challenged some years ago by the paleomagnetic data obtained by Vandenberg (1979) in a small Mesozoic basin located in the Asturias. Surprisingly, the Permo-Triassic redbeds yielded directions apparently very similar to Permo-Triassic data from stable Europe. He attempted to explain this unexpected result by postulating the existence of a micro-plate, called "Le Danois block", which would have behaved independently during the major movement of the Iberian plate coeval with the opening of the Bay of Biscay. The reasons supporting the assigned Permo-Triassic age to the magnetic components were:

(1) Since the NRM behaved as a single compo-

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nent, there was no evidence for any remagnetization.

(2) Both polarities, although not exactly antiparallel, were found.

(3) The dip correction of the monoclinal structure resulted in shallow inclinations consistent with the assumed Permo-Triassic age of magnetization.

As there was little geological evidence supporting Vandenberg's hypothesis, his results called for further investigation. Our aim was mainly (1) to verify or refute Vandenberg's paleomagnetic interpretation using more data, and (2) to delimit the "le Danois block" more closely.

For these purposes, we extended the sampling to the whole Asturian Mesozoic basin and the part of the Cantabric Chain, on both sides of a complex fracture zone stretching from east to west (Figs. 1 and 2), which might represent the southern limit of the micro-plate.

Furthermore, since the paleomagnetic pole position for the upper Permian–lower Triassic is poorly defined for Iberia, the need for new data was also justified.

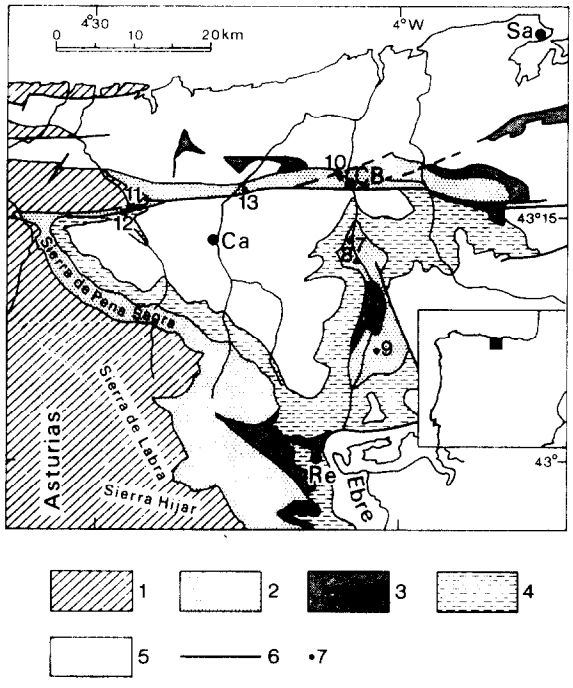


Fig. 1. Geological map of the western part of the Cantabric Chain and localization of the sampling sites. *1* = Paleozoic basement of the Asturias; *2* = Permo-Triassic; *3* = Keuper; *4* = Jurassic; *5* = Cretaceous and Tertiary; *6* = main faults; *7* = sampling sites. *Ca*—Cabuerniga; *CB*—Las Caldas de Besaya; *Re*—Reinosa; *Sa*—Santander.

2. Geological setting and description of the sampling localities

2.1. Cantabric chain (Fig. 1)

The redbeds are exposed on the eastern border of the Paleozoic Asturian massif. They are found discordantly overlying either shales of unknown age or Carboniferous limestone. Saiz de Omenaca (1977) divides the continental deposits into two or three sedimentary cycles depending on locality. In the sampled area, sedimentation apparently began later than in the western basins (Sierras de Pena Sagra, Pena Labra, Hijar) and comprised only two cycles, reaching a thickness of 100 and 260 m respectively. The lower beds consist of conglomeratic sandstone. They pass into finer-grained sandstone and siltstone, red to red-violet in colour. The uppermost layers contain saliferous beds which are more frequently found when the distance from the original sedimentary basins (see above) increases. Magmatic effusives with an ophitic texture are associated with the top of the continental series, but the tectonic correlations with the surrounding rocks are unclear. Evaluating the age of the redbeds is difficult due to the absence of fossiles. The discovery of one specimen of Voltzia has provided some evidence for a lower Triassic age.

The main tectonic feature involving Paleozoic rocks and their Permo-Triassic cover is the so-called Escudo de Cabuerniga anticline stretching E–W over 60 km, and overthrusting Cretaceous formations toward the south. In the kinematic model proposed by Vandenberg, this prominent feature might represent the southern boundary of the "Le Danois block". The whole area was certainly involved in the tectonic events which deformed the Spanish margin during the Eocene and Oligocene. In particular, the E–W running thrust fault is the western extension of similar accidents known in the Asturias which were inherited from the late Variscan stage, and were remobilized during the Oligocene (see next section).

The sampling sites are mainly situated in the

valley of the Besaya river, which offers a good section through the Escudo de Cabuerniga anticline, and through the Permo-Triassic outcrops located about 10 km north of Reinosa (four sites). Three other localities, situated farther west, belong to the Escudo de Cabuerniga anticline.

2.2. Asturias (Fig. 2)

The small Mesozoic basin located in this region is divided into two areas, separated by a complex fracture zone (Julivert et al., 1971). The northern part consists mainly of flat-lying Jurassic limestones, whereas the southern part takes the form of a narrow depression stretching from east to west, which is filled in with Cretaceous and Tertiary formations. The Permo-Triassic redbeds are exposed mainly in the northern area. The lower part of the continental deposits is attributed to the Permian. It begins with a polygenic conglomerate which passes into sandstone, conglomerate and red coloured siltstone. They contain a few intercalations of effusive material. The estimated dating is supported by a flora discovered by Patac (1920) near Pola de Siero, of Autunian age, but unfortunately its stratigraphic localization is rather imprecise. The formations attributed to the Buntsandstein comprise conglomerates at the bottom levels, overlain by finer-grained red to red-violet coloured clastics. The overall thickness reaches 800 m. The top of the Triassic consists of variegated clays and evaporitic facies attributed to the Keuper. The Triassic is overlain by Jurassic sediments found either in marine or Purbeckian facies in the north, and directly by Cretaceous deposits in the south.

The whole Asturian arc is cut up by a complex frame of wrench faults inherited from the Variscan or late Variscan orogeny. In the region under consideration one of the most important structures is the narrow, highly deformed system which separates the Mesozoic basin into the two parts described above. Another notable example is afforded by the Ventaniella fault, NW–SE in direction, which stretches through the whole Asturian massif. This fault system was remobilized several

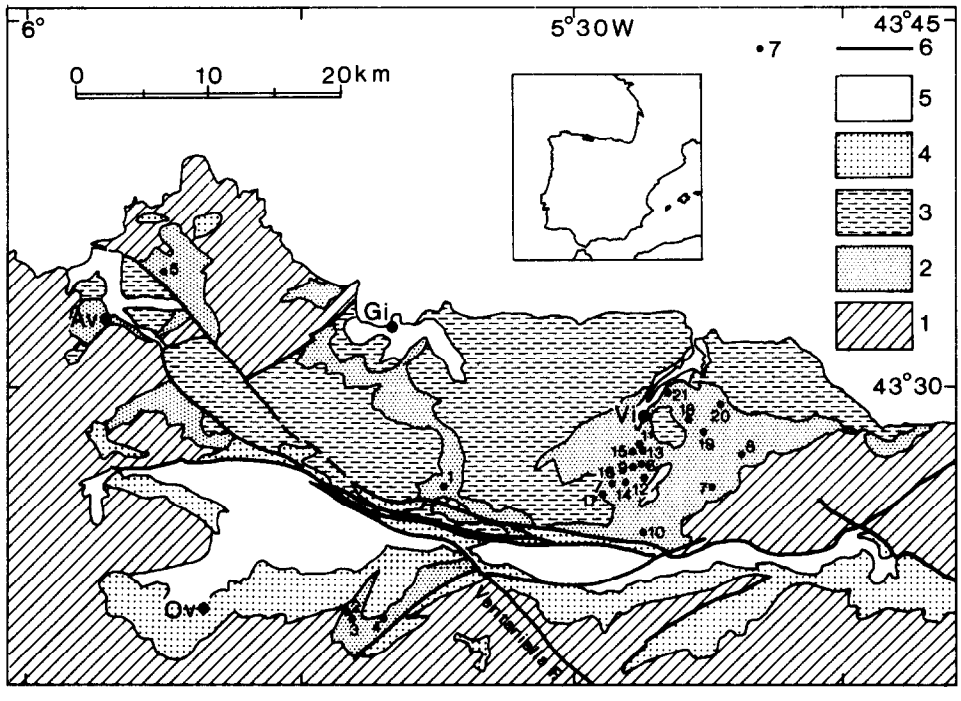


Fig. 2. Geological map of the Mesozoic basin of the Asturias and localization of the sampling sites. *1* = Paleozoic; *2* = Permo-Triassic; *3* = Jurassic; *4* = Cretaceous; *5* = Tertiary; *6* = main faults; *7* = sampling sites. *Av*—Aviles; *Gi*—Gijon; *Ov*—Oviedo; *Vi*—Villaviciosa.

times during the Mesozoic and Tertiary.

One important episode took place in the lower Cretaceous, with vertical movements controlling the sedimentation. The paroxysmal phase occurred in the lower Tertiary and was sealed by the conglomerate of Posada, whose assigned age is lower Oligocene. The age of this prominent event is confirmed by the deformations observed offshore on the continental margin (Boillot et al., 1973). These authors describe two main stages of deformation: a first compressionnal phase in the middle Eocene, and a second episode, dating from the lower Oligocene, and characterized by a slight compression and vertical movements.

The sampling sites are located mainly in the small basin of Villaviciosa where the Permo-Triassic redbeds are well exposed (sixteen sites). The exposures become less apparent towards the east, where only two more outcrops were sampled. Three sites belong to both flanks of a small syncline located in the southern part of the Mesozoic basin, some 15 km west of Oviedo.

3. Sampling and analytical techniques

The sampling was done using a portable gas-powered core drill, and cores were oriented with magnetic and sun compasses and an inclinometer. Whenever possible, the samples were evenly distributed over the outcrop as well as over a stratigraphic thickness as large as possible. The height varied from 0.3 m up to 40 m. Each core was cut into two or three standard core specimens 2.5 cm in diameter and 2.5 cm long. The Natural Remanent Magnetization (NRM) of the specimens was measured with a fluxgate "Digico" magnetometer. At least one specimen per sample was submitted to progressive thermal demagnetization up to 680 ° C.

The susceptibility was measured initially and then once again after each step of treatment with a commercial a-c bridge (Digico) in order to detect the mineralogical changes induced by heat. These effects sometimes gave rise to a notable increase in susceptibility and to the acquisition of spurious magnetizations which prevented us from achieving a complete demagnetization. In addition, chemical demagnetization was carried out on specimens selected from samples with typical thermal demagnetization curves. The procedure is described elsewhere (Schott and Peres, in press). The treatment was performed in a field-free space. In every case, but one (site AS5), hydrochloric acid was employed. In site AS5, however, this technique led to a rapid disaggregation due to the presence of carbonates. We were able to avoid this inconvenience by using a buffer solution containing sodium citrate, sodium bicarbonate and sodium hydrosulfite (Pozzi, pers. commun.). It revealed itself to be less efficient than hydrochloric acid, but a fairly reasonable state of demagnetization was reached nevertheless.

4. Paleomagnetic analysis

The analysis of the NRM was carried out either by stepwise thermal demagnetization or by stepwise acid leaching. The difficulties inherent to the latter method are well known (e.g. Henry, 1979) and the relationship between the grain size and the dissolution rate is not simple (Turner, 1981). The comparison of the demagnetization curves obtained using the two methods in the present study shows that the thermal method is more efficient for the separation of the different magnetic components. We always observe more overlap of the dissolution rates spectra than of the blocking temperatures spectra. Moreover, the hardness of the magnetic components is often in reversed order, so that the magnetic component possessing the highest blocking temperatures is the first one removed by acid leaching. This component is always interpreted as the primary one, and therefore the chemical demagnetization mainly allows us to determine the secondary components. Figure 3 displays an illustration of this general behaviour; Fig. 3a is the diagram of the thermal demagnetization of specimen AS8-6-1. The curves reveal clearly that two components are present in the NRM. Figure 3b shows the result of a combined chemical-thermal demagnetization of specimen AS8-6-2. The first part corresponds to the acid leaching process. The treatment was stopped after 371 hours. The shape of this part of the curves is quite different from that of the Fig. 3a. It can be interpreted as the result of the almost

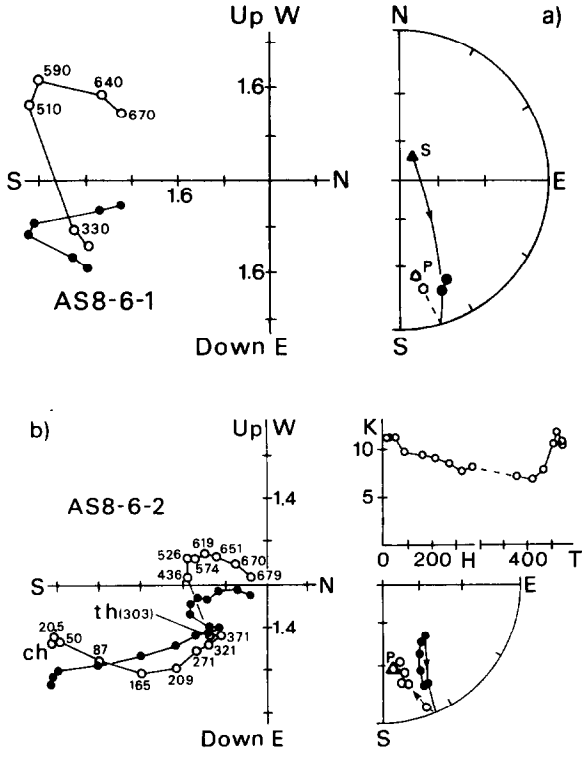


Fig. 3. Comparison of (a) thermal and (b) combined chemical and thermal demagnetizations (the chemical treatment was stopped after 371 h of cumulative leaching). On the orthogonal diagrams solid circles are projections onto the horizontal plane, and open circles are projections onto the vertical N–S plane. Units along the axes are in 10^{-3} A/m. On the Wulff-net plot of the demagnetization paths, solid circles are projections onto the lower hemisphere. Solid triangles represent the directions of the magnetic components. *P*—primary component; *S*—secondary component; *K*—susceptibility per unit volume in 10^{-5} S.I., *H*—cumulative hours of leaching; *T*—temperature. *ch*—chemical leaching; *th*—thermal demagnetization.

simultaneous removal of the two components identified by the former method. If the chemical leaching had been carried on, only the secondary component would have been detected. But the following thermal demagnetization curves show that a non-negligible part of the primary component was still present after the 371 hours of chemical leaching. Its direction can be well deduced from the final portions of the curves. It can also be roughly estimated on the chemical demagnetization diagram, by vectors subtraction between 87 h and 50 h. The vectors subtraction between 436° and 371° yields an estimation of the secondary component. The directions are in accordance with the data based on the thermal demagnetization alone.

The analysis based mainly on the thermal demagnetizations reveals that the NRM's consist of several components, the significance of which will be discussed below. The results will be presented separately for the Cantabric Chain and for the Asturias.

4.1. Cantabric Chain

Two magnetic components are identified in the NRM. The high temperature components are displayed in Fig. 4. We assume that they represent the primary magnetization dating from lower Triassic, on the basis (1) of the fold test of McElhinny (1964), positive at the 99% level, (2) the low inclinations obtained after unfolding and (3) the occurrence of both polarities consistent with the polarity time scale of McElhinny and Burek (1971). Our results confirm the directions found by Vandenberg (1979) in the same area, although the latter display somewhat higher inclinations. The deflection of the declinations from European contemporaneous directions is consistent with the deviation calculated using lower Permian data (Van der Voo, 1969).

This evidence suggests that these primary magnetic components and their corresponding paleomagnetic poles can be considered as characteristic lower Triassic values for the Iberian plate.

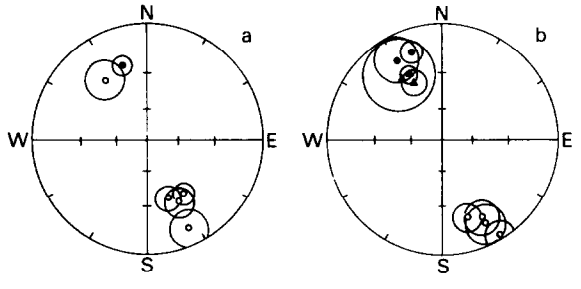


Fig. 4. Stereographic plot of the site mean directions and circles of confidence of the primary components from the Cantabric Chain. Solid circles are projections onto the lower hemisphere, open circles are projections onto the upper hemisphere. Solid triangles are the site mean directions obtained by Vandenberg (1979). a. Before tectonic correction. b. After tectonic correction.

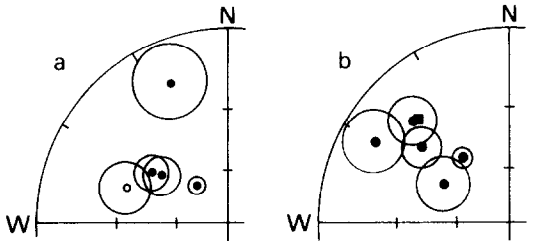


Fig. 5. Stereographic plot of the site mean directions and circles of confidence of the secondary components from the Cantabric Chain. (Both polarities occur, but the reversed components are only scarcely determined and yield no site mean directions). Solid circles are projections onto the lower hemisphere, open circles are projections onto the upper hemisphere. Solid square is the direction calculated from the lower Cretaceous pole of the Wealden series (Schott and Peres, in press). a. Before tectonic correction. b. After tectonic correction.

The secondary components possess blocking temperatures reaching 670 °C. They are present in all sites but one (CT7) and both polarities occur too, although the reversed components have not been accurately determined. The fold test is positive at the 95% level and, therefore, shows that the secondary components also predate the folding. Figure 5 displays the site mean directions yielded by the normal components. The directions obtained after unfolding are quite similar to the primary directions found in the Wealden series from the Sierra de Los Cameros (Schott and Peres, in press). Accordingly, the same lower Cretaceous age will be assigned to this remagnetization phase.

In short, the paleomagnetic analysis of the lower Triassic redbeds from the studied area reveals that the NRM's consist of two components. A primary component, dating from lower Triassic yields a lower Triassic paleopole for the Iberian plate. The original component is overprinted by a secondary magnetization acquired during the lower Cretaceous.

The meaning of the remagnetization phase will be discussed in section 5, together with the results obtained from the Asturias. Tables 1 and 2 summarize the data for both types of paleomagnetic directions.

4.2. *Asturias*

The paleomagnetic analysis mainly based on thermal cleaning revealed itself to be more intricate than in the Cantabric Chain. Four types of magnetic components were identified and will be described below in order of decreasing age:

Group 1. These components were identified in 5 sites out of 21 (see Table 3). They possess the highest blocking temperatures, reaching the Curie temperature of haematite. Only reversed polarities are present (Fig. 6). The fold test is positive at the 90% level only, due to the small number of direc-

TABLE 1

Site mean directions and paleomagnetic poles of the primary and secondary magnetizations from the Cantabric Chain *

Site	n/n_0	Site mean directions				Statistics			Paleomagnetic poles			
		without tect. cor.		with tect. cor.		N	k	α_{95}	lat.°N	long.°E	dp	dm
		D_m	I_m	D_m	I_m							
CT 7, prim. dir.	7/7	341.5	22	341	13	7	79	6.5	50	206	3.5	7.0
CT 8, prim. dir.	12/8	155	−9.5	153	−16	7	28	11.0	48	218	6.0	11.5
CT 9, prim. dir.	9/6	153	−27.5	153	−11.5	3	118	11.5	46	216	6.0	11.5
second. dir.		319.5	62.5	323	46.5	5	308	4.5	57	251	3.5	5.5
CT10, prim. dir.	16/11	160.5	−33.5	162	−19	5	74	9.0	53	206	4.5	9.0
second. dir.		302.5	40	309.5	29.5	8	40	8.5	39	249	5.5	9.5
CT11, second. dir.	4/3	288.5	−32	300	13	4	61	11.5	26	249	6.0	12.0
CT12, prim. dir.	16/10	324.5	−26	330.5	14	6	25	13.5	45	220	7.0	13.5
second. dir.		337	15	299	48	6	24	13.5	40	270	11.5	18.0
CT13, prim. dir.	13/8	146	−31.5	149	−2	4	140	8	39	218	4.0	8.0
second. dir.		304	45	315.5	19.5	7	41	9.5	39	238	5.0	9.5

* n/n_0—number of specimens/number of samples; N—number of directions used in the Fisher statistics; k, α_{95}—parameters of Fisher statistics; dp, dm—semi-angles of the ellipse of confidence. Paleomagnetic poles were calculated after tectonic correction.

TABLE 2

Overall mean directions and paleomagnetic poles of the primary and secondary magnetizations from the Cantabric Chain *

Directions	Mean directions				Statistics					Mean paleom. pole			
	without tect. cor.		with tect. cor.		without t.c.			with t.c.		lat.°N	long.°E	k	α_{95}
	D_m	I_m	D_m	I_m	N	k	α_{95}	k	α_{95}				
Primary	333.5	17	334.5	12.5	6	12	20	107	6.5	47	214.5	161	5.5
Secondary	310.5	29	309	31.5	5	4.3	41.5	21	17	40.5	251.5	33	13.5

* (Primary components are all converted into normal polarities.) Paleomagnetic poles were calculated after tectonic corrections. Statistical parameters as in Table 1.

TABLE 3

Site mean directions and paleomagnetic poles of group 1 (original magnetization) from the Asturias *

Site	n/n_0	Site mean directions				Statistics			Paleomagnetic poles			
		without tect. cor.		with tect. cor.		N	k	α_{95}	lat.°N	long.°E	dp	dm
		D_m	I_m	D_m	I_m							
AS2	9/6	153	−10	155.5	−19	9	99	5.0	50	214	3.0	5.5
AS3	8/6	153	10	153.5	−17	9	100	5.0	48	216	2.5	5.5
AS7	8/5	164.5	1.5	164	−10.5	8	346	3.0	49	200	1.5	3.0
AS8	9/6	164	−30.5	170.5	−16	8	122	5.0	54	191	2.5	5.0
AS10	9/4	171.5	−16.5	172.5	−7	8	176	4.0	50	186	2.0	4.0

* Paleomagnetic poles were calculated after tectonic corrections. Statistical parameters as in Table 1.

tions and to the fact that the tectonic corrections have little influence. After unfolding, the directions are consistent with the similar data from the Cantabric Chain, although the main directions from sites 7, 8 and 10—belonging to the area located north of the E–W striking fracture zone—display a deviation in declination of 15° with respect to the mean value yielded by sites 2 and 3 (situated south of the fracture zone) and the lower Triassic directions from the Cantabric Chain. The use of the statistical test of McFadden and Lowes (1981) shows that the angular deviation between the two mean directions is significant at the 95% level.

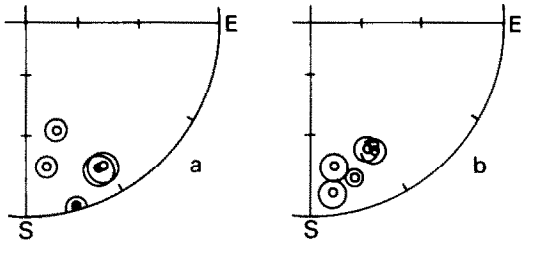


Fig. 6. Stereographic plot of the site mean directions and circles of confidence of group 1 from the Asturias. Solid circles are projections onto the lower hemisphere, open circles are projections onto the upper hemisphere. Open square is the direction calculated with the Permo-Triassic pole from the Cantabric Chain. a. Before tectonic correction. b. After tectonic correction.

Group 2. The components classified in group 2 occasionally appear in 6 sites from the basin of Villaviciosa (Table 4). Both polarities occur (Fig. 7). The blocking temperatures spectra of these components is highly variable. In some cases it ranges from 180° up to 640° but in a few other cases, it covers a narrower interval, from about 520° to 640°. The fold test is positive at the 95% level. Unfolding brings the directions to a location very similar to that of the secondary components from the Cantabric Chain. As in that case, the mean is very close to the direction deduced from the lower Cretaceous paleomagnetic pole yielded by the Wealden series from the Sierra de Los Cameros. Therefore we ascribe a lower Cretaceous age to the magnetization represented by group 2.

Group 3. When they are present, these compo-

TABLE 4

Overall mean direction and paleomagnetic pole of group 2 (lower Cretaceous remagnetization phase) from the Asturias *

Site [a]	n [b]	Mean directions				Statistics					Mean paleom. pole			
		without tect. cor.		with tect. cor.		without t.c.			with t.c.		lat.° N	long.° E	dp	dm
		D_m	I_m	D_m	I_m	N	k	α_{95}	k	α_{95}				
AS1, AS7, AS8, AS10, AS17, AS21	15	316	35	318	21.5	11	14	12.0	34	8.0	42	236	4.5	8.0

* The paleomagnetic pole was calculated after tectonic corrections.

[a] Sites where the remagnetization was identified.

[b] Total number of sample directions (both polarities are taken into account). Other symbols are in Table 1.

nents are readily identified: in this case the NRM's behave as single components and show a tight grouping within each site. The directions are not significantly shifted by thermal cleaning (Fig. 8). They correspond to rather soft magnetic components, characterized by blocking temperatures not exceeding 620° and by fast removal using chemical demagnetization.

In the siltstone and fine sandstone, the decrease of the NRM's is further enhanced by the existence of small cavities filled in with carbonates. Their rapid dissolution notably increases the leaching surface of the samples. Figure 8 shows that the site mean directions are more scattered after tectonic correction. The fold test of McElhinny (1964) is negative at the 95% level. The dispersion is not random, the directions in fact being approximatively distributed along a portion of a great circle. This feature merely reflects the structural attitude of the beds: they are deformed in a slightly folded monocline, with fold axes striking NW–SE, as can be seen in Fig. 8b, from the normals to the bedding planes. Three reversed site mean directions are classified into group 3: the reversed directions mentioned by VandenBerg (1979), labelled VDB in Fig. 8, and the directions of sites AS5 and AS4. The significance of the data from site AS4 is questionable, as this data could

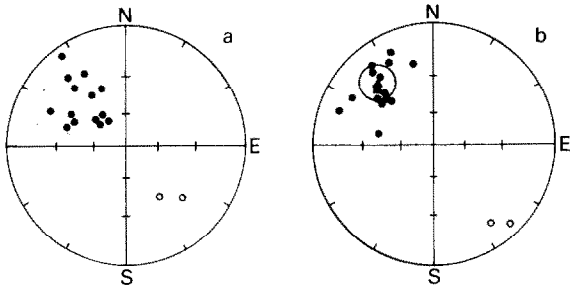


Fig. 7. Stereographic plot of the individual directions of group 2 from the Asturias. Solid circles are projections onto the lower hemisphere. Open circles are projections onto the upper hemisphere. Solid triangle is the mean direction (after tectonic correction). Solid square is the direction calculated from the lower Cretaceous pole of the Wealden series (Schott and Peres, in press). a. Before tectonic correction. b. After tectonic correction.

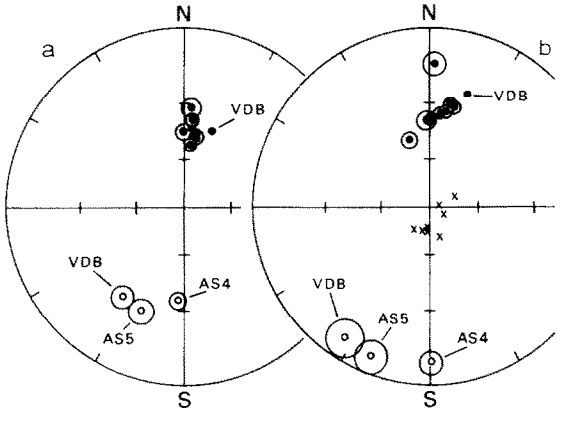


Fig. 8. Stereographic plot of the site mean directions and circles of confidence of group 3 from the Asturias. Solid circles are projections onto the lower hemisphere. Open circles are projections onto the upper hemisphere. Crosses are the normals to the bedding planes. VDB—directions obtained by Vandenberg (1979), (because normal directions are very tightly grouped the small circle of confidence has not been drawn). See text for discussion about site AS4. a. Before tectonic correction. b. After tectonic correction.

also be attributed to group 1. We will discuss this point further in the next section. In addition to the reversed directions just cited, we mention reverse components detected in sites AS 18, 19, 20, 21, where they are superimposed upon components belonging to groups 3 or 4 and possess the highest blocking temperatures. But the overlap of the blocking temperatures does not allow us to determine them exactly. Their common mean can be estimated from the convergence circles (Halls, 1976) by the statistical method of Bailey and Halls (1984), whose result is given in Table 5. The magnetic components of group 3 are clearly post tectonic, but it is difficult to estimate an age of magnetization more precisely than lower Tertiary. This assumption is based on the relative low values of inclination, a fact which is noted by Westphal et al. (1986) for data attributed to the Eocene.

Group 4. This last category merely groups the components of recent origin. They are characterized by a wide spectrum of blocking temperatures and high resistance to chemical cleaning, as was illustrated by Fig. 3. Only normal polarities occur (Fig. 9 and Table 6). The fold test is negative, demonstrating the post-tectonic acquisition

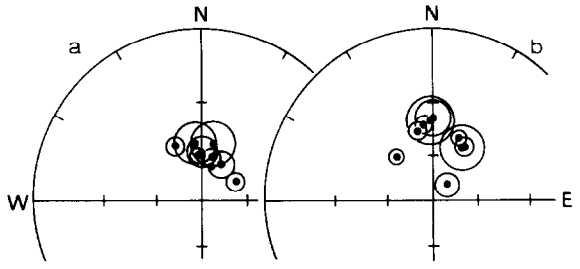


Fig. 9. Stereographic plot of the site mean directions and circles of confidence of group 4 from the Asturias. Projections are onto the lower hemisphere. a. Before tectonic correction. b. After tectonic correction.

of these components. The location of their mean as well as their polarities are consistent with the assumption of a recent acquisition.

Table 7 summarizes the mean directions and poles of the four types of components identified in the Permo-Triassic redbeds from the Asturias.

5. Discussion

Taking into account both areas studied, the paleomagnetic analysis shows that the NRM of

TABLE 5

Site mean directions and paleomagnetic poles of group 3 (lower Tertiary remagnetization phase) from the Asturias *

Site	n/n_0	Site mean directions				Statistics			Paleomagnetic poles			
		without tect. cor.		with tect. cor.		N	k	α_{95}	lat. °N	long. °E	dp	dm
		D_m	I_m	D_m	I_m							
AS4	6/4	183.5	−35	179.5	−8.5	7	214	4.0	66	167	2.5	4.5
AS5	12/5	202.5	−26	201.5	−6.5	11	61	6.0	54	134	3.5	6.0
AS6	10/9	10	45.3	11.5	29.5	10	217	3.0	72	145	2.5	4.0
AS9	9/8	10.5	46.5	10	33.5	9	233	3.5	72	143	2.5	4.5
AS11	9/8	8.5	47	14	30.5	9	230	3.5	73	148	3.0	4.5
AS12	8/7	4.5	32	2	13	9	134	4.5	64	165	3.0	5.0
AS13	8/6	5.5	51.5	6.5	34.5	8	310	3.0	78	151	3.0	4.0
AS14	14/10	8.5	19	–	–	14	33	7.0	56	160	3.5	7.0
AS15	10/9	5.5	43	358	38.5	12	95	4.5	71	159	3.5	5.5
AS16	9/8	5.5	37.5	1	38	8	373	3.0	67	161	2.0	3.5
AS18	13/12	359.5	44.5	343	47.5	13	103	4.0	73	175	3.0	5.0
Conv. Circles AS18, AS19, AS20 [a]		185	−32	–	–	13	161 [b]	$l_1 = 3.5$ $l_2 = 14.5$	63	164	–	–

* Paleomagnetic poles were calculated without tectonic corrections.

[a] List of the sites where reversed components are identified by the method of converging remagnetizing circles.

[b] l_1, l_2—semi-angles of the ellipse of confidence (Bailey and Halls, 1984). Other symbols as in Table 1.

TABLE 6

Site mean directions and paleomagnetic poles of group 4 (recent remagnetization phase) from the Asturias *

Site	n/n_0	Site mean directions				Statistics			Paleomagnetic poles			
		without tect. cor.		with tect. cor.		N	k	α_{95}	lat. °N	long. °E	dp	dm
		D_m	I_m	D_m	I_m							
AS1	9/9	1	58	0.5	38	5	66	9.5	85	163	10.5	14.0
AS2	9/6	11.5	52.5	29	51	3	88	13.0	76	129	12.5	18.0
AS5	12/5	353.5	52.5	358.5	39.5	5	39	12.5	78	201	11.5	17.0
AS7	8/5	28.5	62.5	42.5	75.5	6	68	8.0	69	74	10.0	12.5
AS8	9/6	16.5	66	23	46	7	197	4.0	77	55	5.5	7.0
AS17	19/10	14	60	348	44.5	14	64	5.0	79	93	5.5	7.5
AS19	11/9	355	60	321	52.5	12	111	4.0	86	231	4.5	6.0
AS20	8/7	334	50.5	353.5	41.5	8	122	5.0	66	242	4.5	6.5
AS21	19/10	60	64	31	49	19	35	5.5	48	60	7.0	9.0

* Paleomagnetic poles were calculated without tectonic corrections. Symbols as in Table 1.

TABLE 7

Overall mean directions and paleomagnetic poles of the 4 groups of magnetic components carried by the redbeds from the Asturias *

Directions	Mean directions				Statistics					Mean paleom. pole			
	without tect. cor.		with tect. cor.		without t.c.			with t.c.		lat. °N	long. °E	k	α_{95}
	D_m	I_m	D_m	I_m	N	k	α_{95}	k	α_{95}				
Group 1	161	−9	163.5	−14	5	21	17	70	9	51	201.5	84	8.5
Group 2	316	35	318	21.5	11	14	12.0	34	8.0	42	236	–	–
Group 3	6	43.5	4	33.5	8	152	4.5	42	8.5	71.5	156.5	221	3.5
Group 4	8.5	61.5	5	50.5	9	43	8	18.5	12.5	83	96.5	18	12.5

* Paleomagnetic poles were calculated after tectonic corrections for groups 1 and 2, before tectonic corrections for groups 3 and 4. For group 3, only normal polarities were used in the statistics. Statistical parameters as in Table 1.

the Permo-Triassic redbeds are multi-component vectors. A primary component, classified into group 1, is found in 6 sites out of 7 in the Cantabric Chain and in 6 sites out of 21 in the Asturias. It displays directions consistent with a Permo-Triassic age of magnetization. Mixed polarities are present in the Cantabric Chain, but only reversed polarities in the Asturias. The deviation in declination from European contemporaneous data is similar to the value obtained by Van der Voo (1969) from lower Permian results. Hence, directions and paleomagnetic poles yielded by group 1 are typical Permo-Triassic data for the Iberian plate. This interpretation does not support the hypothesis of Vandenberg (1979) about the existence of the "le Danois" micro-plate. This author postulated the existence of this unit in order to explain the occurrence of apparently European directions in the small basin of Villaviciosa. In fact, the directions obtained by Vandenberg are, in our opinion, not of primary origin but were actually built up during a remagnetization phase classified into group 3. Nevertheless, a small difference of 15° can be observed between the declinations obtained north of the narrow fracture zone cutting the Mesozoic basin of the Asturias into two parts (see section 2), and the remaining data. This deviation may be explained by a slight clockwise rotation of this area, as can be observed in other parts of the boundary between the European and the Iberian plates (Schott, 1985), and can be interpreted merely as a local effect induced by the convergence of the plates during the early Tertiary.

The primary component is overprinted by 3 remagnetization phases of unequal importance. The older phase gave rise to magnetic components classified into group 2. In the Cantabric chain these components were recognized in 5 sites out of 7 (Table 1), which allowed for the calculation of site mean directions, whereas in the Asturias, this sort of component is more rare and allows only for the calculation of a mean direction based on individual values (Table 4). We assign an upper Jurassic–lower Cretaceous age to this remagnetization phase owing to the similarity between these directions and polarities and the data obtained on the lower Cretaceous Wealden series from the Sierra de Los Cameros (Schott and Peres, 1987), and on upper Jurassic limestone from the Iberian Cordillera (Steiner et al., 1986). The occurrence of these anomalous directions, accounting for a "hairpin" in the Iberian apparent polar wander path at that time, is discussed by Schott and Peres. The study of Steiner et al. shows that the anomalous behavior apparently extends back to the upper Jurassic and consequently has a greater duration than supposed by Schott and Peres. As we discussed in that article, this first remagnetization phase may be connected with the distensive phase that began in the Pyrenean and Cantabrian areas in the upper Jurassic (Salomon, 1982; Malod, 1982).

The second remagnetization phase represented by the directions of group 3, very extensively affected the redbeds of the Asturias and resulted in an almost complete replacement of the original magnetization. The number of site mean directions is large enough to ensure that the fold test is negative at the 95% level despite the small influence of the tectonic corrections, and to demonstrate that these components cannot represent the original magnetization. The shallow inclinations displayed by one end of the cluster obtained after unfolding, found both by us and by Vandenberg (1979), can obviously lead one to erroneously consider them as European directions. Taking into account the directions of group 1, it would be very difficult to explain the occurrence of both European and Iberian directions in the small basin of Villaviciosa.

We also experienced some difficulty interpreting some directions attributed to group 3 in our study, as illustrated by the results from site AS4 (Fig. 8) which display reversed polarities. Their proposed classification in group 3 rather than in group 1 can be a point of disagreement. Although the fold test (performed on reversed polarities only) is not conclusive due to the small amount of data, we feel that the mean direction agrees better with those of group 3 (without tectonic correction), than with the primary directions (with tectonic correction). The discrepancy with the other directions obtained from the same area, south of the E–W fracture zone (namely sites AS2 and AS3), is significant and difficult to explain. Moreover, there is some field evidence supporting the assumption of remagnetization: in site AS4, the redbeds lie just under Albian unconsolidated deposits. This may have facilitated the circulation of fluids. The NRM is nearly a single component, like the NRM's from group 3 and has the same behavior with respect to thermal demagnetization. Finally, the concurrence of the site mean direction of site AS4 and the reversed mean direction revealed in the sites mentioned in section 4.2. by the convergence circles (Table 5), is also in accordance with our interpretation. As the directions of group 3 postdate the folding, we postulate an Eocene age for the inferred remagnetization phase. Our assumption is based on the values of inclinations: such rather low values are found in Eocene formations in several areas around the Mediterranean, as noted by Westphal et al. (1986). Thus, this remagnetization phase is connected with the compressional phase induced along the northern boundary of the Iberian plate by the convergence of Europe and Iberia, beginning in the upper Cretaceous and giving rise to the orogeny.

Lastly, the fourth group of components is interpreted as the result of recent remagnetization. They are characterized by a surprising resistance to chemical leaching and are therefore always well determined by this method. The distinction between group 3 and 4 is somewhat arbitrary in some cases. It is mainly based on the values of inclinations which are higher in group 4 than in group 3.

As there is no field evidence for heating of the sediments, the various remagnetizations described

above are most probably of chemical origin. This process, involving circulation of interstitial fluids and precipitation of authigenic magnetic minerals is well known in redbeds (see Walker et al., 1981, for a review). In the case studied deformations induced by the distensive phase in the lower Cretaceous or by the compressional phase in the Eocene, may have facilitated the circulation of groundwater and the remobilization of iron oxides.

Conclusion

The paleomagnetic study of the Permo-Triassic redbeds from the Cantabric Chain and the Asturias leads us to three main conclusions:

(1) The magnetic history of these formations is much more intricate than expected from previous studies. Apart from recent weathering, two main phases of remagnetization have been found to be superimposed upon the original magnetization. The remagnetization process is variable, ranging from partial overprinting to complete replacement. Each of the remagnetization phases can be associated with a major tectonic event: the oldest phase corresponds to the distension that affected the area in the upper Jurassic–lower Cretaceous, just prior to the first movement of Iberia. The subsequent phase, attributed to the Eocene, is related to the compressionnal phase resulting from the convergence of the European and Iberian plates.

(2) The original magnetization was fortunately preserved in 6 out of 7 sites in the Cantabric Chain, but only in 6 out of 21 sites in the Asturias, demonstrating the widespread nature of the remagnetization processes in this area. The data from the Cantabric Chain, along with the results given by Vandenberg (1979) yield a reliable upper Permian–lower Triassic paleomagnetic pole for Iberia. In the Asturias, a distinction may be drawn between the sites located north or south of the fracture zone which divides the Mesozoic basin into two parts: the data which comes from the southern part agrees with the directions from the Cantabric Chain and can be regarded as characteristic directions for Iberia too. The declinations found in the northern part display a deviation of 15°, which may be explained by a slight clockwise rotation of this area, a local disturbance probably induced by the Tertiary compression.

(3) Our results do not support the hypothesis of a micro-plate named "Le Danois block" suggested by Vandenberg (1979). Furthermore, our study shows that the apparent European directions found by that author in fact represent a remagnetization phase occurring in the lower Tertiary. As Vandenberg's suggestion was based on this misinterpreted data, the existence of a microplate cannot be further advocated.

(4) The Permo-Triassic redbeds from the Asturias and to a lesser extent from the Cantabric chain, offer a good example of how intricate the magnetic history of redbeds can be. But, at the same time, their paleomagnetic analysis shows that a careful study of a representative collection of samples can result in the successful identification of the different magnetic components. The main difficulty lies in the estimation of their ages. In the case studied, the magnetic history of the redbeds is closely connected to the succession of tectonic events although these never induced strong deformations.

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